

Word-trace dominance does not imply fractional Schatten-norm dominance

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Abstract

The first question in the concluding remarks of Alhejji and Knill asks whether dominance of all word traces, together with matched individual spectra, forces dominance of all Schatten norms of the sums. A diagonal three-dimensional example answers this question negatively for every order strictly between one and two. We give the complete all-word argument, prove that dimension three is minimal, and check a second example whose decisive comparison uses integers only. The result concerns the proposed implication, not the spin-alignment conjecture itself.

1 The question and a small counterexample

For a positive semidefinite matrix X , write $\|X\|_p = (\operatorname{tr} X^p)^{1/p}$, where $1 \leq p < \infty$. Section VI, question 1 of [1] considers four positive semidefinite matrices with $\lambda(A_i) = \lambda(B_i)$ for $i = 0, 1$. For a finite binary word s , let $\Pi_s(A)$ be the ordered product of the matrices selected by its letters, and define $\Pi_s(B)$ similarly. The question is whether

$$\operatorname{tr} \Pi_s(A) \geq |\operatorname{tr} \Pi_s(B)| \quad \text{for every } s \quad (1)$$

implies $\|A_0 + A_1\|_p \geq \|B_0 + B_1\|_p$ for every $p \geq 1$.

Theorem 1. *Set*

$$\begin{aligned} A_0 &= \operatorname{diag}(2, 0, 1), & A_1 &= \operatorname{diag}(2, 1, 0), \\ B_0 &= \operatorname{diag}(1, 2, 0), & B_1 &= \operatorname{diag}(2, 1, 0). \end{aligned} \quad (2)$$

The matrices have matched individual spectra and satisfy (1). Nevertheless,

$$\|A_0 + A_1\|_p < \|B_0 + B_1\|_p \quad (1 < p < 2). \quad (3)$$

Equality holds at $p = 1, 2$, and the inequality reverses for $p > 2$.

Proof. All four matrices are nonnegative and diagonal, with spectrum $(2, 1, 0)$. A word containing only one letter has the same trace on both sides. This includes the empty word, whose product is the identity. For a mixed word with $k \geq 1$ zeros and $m \geq 1$ ones, commutativity gives

$$\operatorname{tr} \Pi_s(A) = 2^{k+m}, \quad \operatorname{tr} \Pi_s(B) = 2^m + 2^k \leq 2^{k+m}. \quad (4)$$

The last inequality follows from $2^{-k} + 2^{-m} \leq 1$. The traces are nonnegative, so the absolute value in (1) causes no change. This proves the hypothesis for all lengths, without a truncation argument.

The two sums are $\text{diag}(4, 1, 1)$ and $\text{diag}(3, 3, 0)$. Their p -th power traces differ by $4^p + 2 - 2 \cdot 3^p$. After division by $3^p > 0$, the sign is that of

$$g(p) = (4/3)^p + 2(1/3)^p - 2. \quad (5)$$

This function is strictly convex, since $g''(p) = (4/3)^p \log^2(4/3) + 2(1/3)^p \log^2(1/3) > 0$. As $g(1) = g(2) = 0$, strict convexity gives $g(p) < 0$ for $1 < p < 2$. Increasing secant slopes give $g(p) > 0$ for $p > 2$. Taking positive p -th roots proves the claims. At $p = 3/2$, the comparison is $10 < 6\sqrt{3}$, equivalently $100 < 108$. $\square$

2 Why dimension three is minimal

Proposition 2. *In dimensions one and two, the stated hypotheses imply Schatten-norm dominance for every $1 \leq p < \infty$, even when the matrices do not commute.*

Proof. Dimension one is immediate from the matched spectra. In dimension two, put $S_A = A_0 + A_1$ and $S_B = B_0 + B_1$. Matched individual spectra imply $\text{tr } S_A = \text{tr } S_B =: T$ and $\text{tr } A_0^2 + \text{tr } A_1^2 = \text{tr } B_0^2 + \text{tr } B_1^2$. The word 01 gives $\text{tr } A_0 A_1 \geq |\text{tr } B_0 B_1| = \text{tr } B_0 B_1$; the last equality follows because the trace of a product of positive semidefinite matrices is real and nonnegative. Consequently

$$\text{tr } S_A^2 - \text{tr } S_B^2 = 2(\text{tr } A_0 A_1 - \text{tr } B_0 B_1) \geq 0. \quad (6)$$

Write the eigenvalues of S_A and S_B as $T/2 \pm a$ and $T/2 \pm b$, respectively, with $0 \leq a, b \leq T/2$. Equation (6) implies $a \geq b$. For $p > 1$, the function $h_p(x) = (T/2 + x)^p + (T/2 - x)^p$ is nondecreasing on $[0, T/2]$, because its derivative is $p[(T/2 + x)^{p-1} - (T/2 - x)^{p-1}] \geq 0$. Thus $\text{tr } S_A^p \geq \text{tr } S_B^p$. For $p = 1$ equality holds. The case $T = 0$ consists of zero matrices. Theorem 1 therefore attains the smallest possible dimension. $\square$

3 The earlier integer certificate

The matrices displayed in the earlier website note also give a complete counterexample:

$$\begin{aligned} A_0 &= \text{diag}(176, 0, 64), & A_1 &= \text{diag}(80, 49, 0), \\ B_0 &= \text{diag}(64, 176, 0), & B_1 &= A_1. \end{aligned} \quad (7)$$

For mixed words, the ratio of the two traces is

$$\frac{\text{tr } \Pi_s(B)}{\text{tr } \Pi_s(A)} = \left(\frac{64}{176}\right)^k + \left(\frac{49}{80}\right)^m \leq \frac{859}{880} < 1 \quad (k, m \geq 1). \quad (8)$$

Pure words and the empty word again give equality. The sums have diagonals $(256, 49, 64)$ and $(144, 225, 0)$, so

$$\text{tr } S_A^{3/2} = 4096 + 343 + 512 = 4951 < 5103 = 1728 + 3375 = \text{tr } S_B^{3/2}. \quad (9)$$

Every hypothesis and the strict violation are therefore established exactly. No additional kernel lemma is needed for either displayed witness.

Scope, verification, and revision history

Both witnesses were already present in the earlier note dated 17 June 2026. This revision supplies explicit proofs for all words, the full failure interval of the small witness, and dimension minimality. It replaces the earlier statement that a kernel lemma still had to be proved for the displayed diagonal construction. It makes no claim about the earlier, incompletely documented noncommuting example or about a more general diagonal family.

The companion Python script checks the spectra and exact constants, prints the decisive comparisons $100 < 108$ and $4951 < 5103$, and additionally verifies every binary word of length at most 12 for both witnesses: 8,191 words per witness including the empty word. The finite enumeration is a consistency check; the universal conclusions follow from the proofs above. The result does not refute the spin-alignment conjecture or determine a quantum channel capacity. Historical priority and independent expert confirmation remain unverified.

The revision, source comparison, proofs, and checks were prepared with AI assistance under the author’s direction. No independent human mathematical review is asserted. Scientific Agent Skills [2] guided evidence tracking and exposition; it is not a source of mathematical verification.

References

- [1] M. A. Alhejji and E. Knill, *Towards a resolution of the spin alignment problem*, Communications in Mathematical Physics **405**, 119 (2024). doi:10.1007/s00220-024-04980-1. Source statement: [arXiv:2307.06894v3](#), Section VI, question 1.
- [2] T. Kassis, V. Agarwal, Y. He, D. Patel, and A. M. Brueckner, *Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents* (2026). doi:10.48550/arXiv.2609.00065.